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23CSE452 Business Analytics

Case Study

Premium Positioning Viability Predictor - Coimbatore Restaurants

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1. Problem Statement and Objectives

Coimbatore has a large and fast-growing food-service market, ranging from traditional messes and tiffin centres to cafés, bakeries and fine-dining restaurants. For most of these venues the Google Maps listing is the first point of contact with a customer: before visiting, a diner usually compares the star rating, the number of reviews, the price band and the photographs of nearby options. Evidence from review platforms shows that small rating differences change demand. A one-star increase on Yelp was associated with a 5–9% rise in revenue for independent restaurants [1], and an extra half-star made restaurants sell out 19 percentage points more often [2].

This study treats a Google rating of **4.2 stars or more** as the benchmark for *premium positioning*. Below this level, customers are less willing to pay a premium and aggregator platforms give the listing less prominence. Restaurant owners, however, have little evidence on which of the attributes under their control are associated with reaching the benchmark, so investment often goes into interiors and décor while low-cost changes to the online profile, payment options or operating hours are overlooked.

Business problem: *Which attributes that a restaurant owner controls are associated with a Google rating of at least 4.2, and can the chance of reaching this benchmark be predicted from a venue's public listing?* The answer matters to **restaurant owners** planning improvements, to **investors** entering the Coimbatore market and to **food-delivery platforms** that onboard and rank restaurants.

The study has three objectives:

1. To build an original dataset of Coimbatore food-service venues by web scraping public Google Maps listings.
2. To predict whether a venue reaches a rating of 4.2 or higher using only attributes that do not reveal the rating, comparing Logistic Regression, Decision Tree and K-Nearest Neighbours.
3. To measure the effect of each attribute an owner can control and turn the results into ranked, practical recommendations.

2. Data Collection and Dataset Description

No public dataset describes Coimbatore restaurants at this level of detail, so the data were collected by **web scraping Google Maps**, a publicly accessible website [3]. Ready-made datasets from Kaggle, UCI, GitHub or similar repositories were not used.

2.1 Source and collection procedure

Each Google Maps listing (place page) shows the business name, category, address, coordinates, rating, number of reviews, price band, website, opening hours and an *About* section of service, amenity and payment options. A single Google Maps search returns at most about 120 places, so a broad query such as “restaurants in Coimbatore” reaches only a small part of the market. The district was therefore covered with many narrow searches, each combining a food keyword with a locality, and the results were merged afterwards. Table 1 summarises the procedure.

Table 1: Data collection summary.

Item	Detail
Website	Google Maps public business listings for Coimbatore district, accessed without login [3]
Search design	20 food keywords (e.g. restaurant, cafe, bakery, biryani, mess, tea shop, juice shop) × 60 localities from R.S. Puram and Gandhipuram to Pollachi and Mettupalayam = 1,200 searches, e.g. “biryani in RS Puram, Coimbatore”
Tool	Open-source Google Maps scraper by gosom [4], run in Docker; a headless Chromium browser scrolls each result list and opens every place page

Item	Detail
Collection date	22–23 September 2026; stopped at about 4,900 unique places, when new searches mostly returned places already collected
Raw output	4,917 records with 36 fields, several stored as nested JSON (opening hours, About attributes, owner details, sample reviews)
Pilot	200 listings collected earlier with the Apify Google Places Crawler [5]; its free credit could not cover the district, so the pilot is not used in the analysis
Privacy	Only public business data; no login or anti-bot measure bypassed; reviewer names, review text, photos, owner IDs, e-mails and phone numbers removed

2.2 Dataset construction

The raw records were reduced to the final modelling sample in seven steps (Figure 1). The first four build the collected dataset `raw_gmaps_scrape.csv`: duplicates returned by several searches were removed using Google's unique `place_id`; each of the 167 Google categories was mapped to one of eight restaurant types, removing 236 places outside food service such as grocery stores, wholesalers and lodges; and 5 places outside the district boundary were dropped. The last three steps, explained in Section 3.1, prepare the data for modelling.

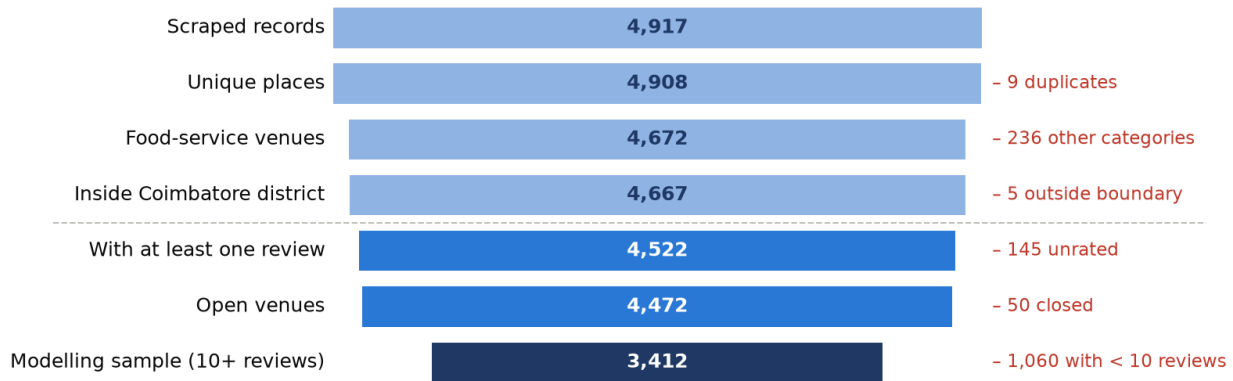


Figure 1: Records remaining after each collection and preparation step (light bars: collected dataset; darker bars: modelling sample).

2.3 Dataset description

The collected dataset contains **4,667 venues** and **29 attributes** in four groups: identity and location (name, Google category, restaurant type, address, area, town, pin code, latitude, longitude); reputation (rating, number of reviews, number of 1- to 5-star reviews); commercial details (price range, website, phone flag, claimed-profile flag, closed flag, online-ordering and table-booking partners, menu link); and operations (opening hours, popular times and the *About* attributes). Table 2 shows sample records.

Table 2: Sample records from the collected dataset.

Business name	Google category	Area	Rating	Reviews	Price	Claimed	Website
Chinese Wok	Chinese restaurant	Hope College	4.1	388	₹200–400	Yes	Yes
Annapoorna - Rajastreet	Vegetarian restaurant	Town Hall	4.0	3,763	₹1–200	Yes	No
Black Door Cafe	Cafe	Hopes	4.7	220	–	No	No
Denmark Restaurant	Chettinad restaurant	Ganapathy	4.0	1,188	₹200–400	No	No
Cream Stone	Ice cream shop	Saibaba Colony	4.8	297	–	Yes	Yes
A1 Mess	Biryani restaurant	Sulur	4.1	958	₹200–400	Yes	No

Multi-cuisine restaurants (28%), bakeries and dessert shops (24%) and cafés (19%) make up most of the collected venues (Figure 2a). The target high_rating equals 1 when the rating is 4.2 or higher. Ratings in the modelling sample are concentrated between 3.8 and 5.0, and **60.9%** of venues reach the benchmark, a moderate class imbalance (Figure 2b).

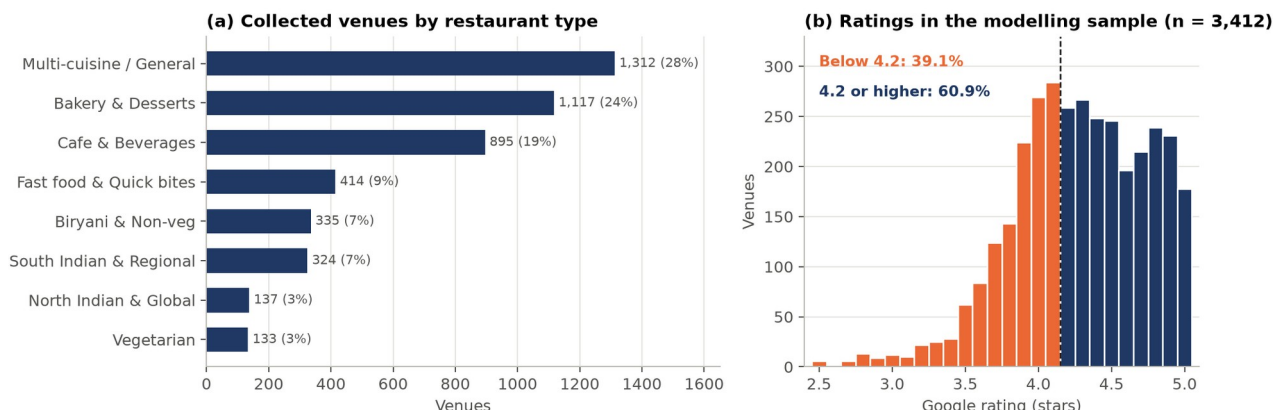


Figure 2: (a) Collected venues by restaurant type; (b) rating distribution of the modelling sample with the 4.2 benchmark (11 venues rated below 2.4 are outside the plotted range).

3. Data Preparation and Exploratory Analysis

3.1 Cleaning and preprocessing

Row selection. Three rules define the modelling sample (Figure 1). Venues without reviews (145) have no real rating, since Google shows them as 0, and were excluded; venues marked as closed (50) are no longer in the market; and venues with fewer than 10 reviews (1,060) were excluded because their ratings are inflated and noisy (Figure 3). Venues with 1–4 reviews reach 4.2 in 78% of cases, compared with 50% for venues with more than 500 reviews, and the spread of ratings falls from 0.78 to 0.33 stars. A handful of favourable early reviews can lift a new venue above the benchmark by chance, so the review count is also kept in the models as a control.

Why venues with fewer than 10 reviews are excluded (grey bars)

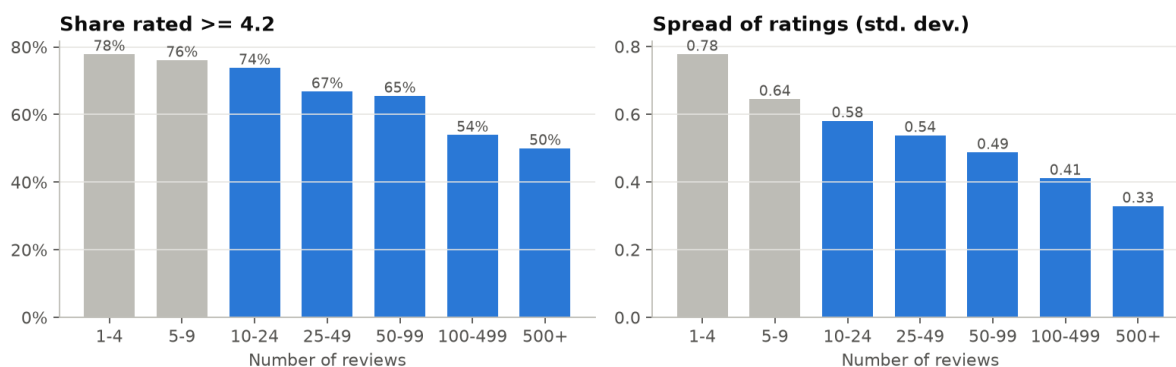


Figure 3: (a) Share of venues rated 4.2 or higher and (b) spread of ratings by number of reviews; grey bars are excluded from modelling.

Field transformations. Most raw fields cannot be used directly: several are nested JSON, some are free text and many are partly missing. Table 3 lists how each field was converted into model features. Missing values were treated as information rather than deleted, which keeps all 3,412 venues in the analysis. In total, 29 features were created (27 numeric or yes/no and 2 categorical), giving 37 model columns after one-hot encoding.

Table 3: Transformation of raw fields into model features.

Raw field	Issue	Transformation	Resulting features
Google category	167 distinct categories	Mapped to 8 restaurant types; brand names appearing 3 or more times marked as chains	concept, is_chain
Price range	Missing for 63.8% of venues; mixed ranges such as ₹1–200, ₹200–1,000	Upper bound grouped into Budget ($\leq$ ₹200), Mid ($\leq$ ₹400), Premium; missing kept as <i>Not disclosed</i>	price_tier
About attributes	Nested JSON; missing for 32.6%	Flattened; an option counts only when marked enabled; missing section flagged	14 yes/no flags, has_attributes
Opening hours	Text such as “11–3 pm”, “7 pm–2 am”, “Open 24 hours”; missing for 10.9%	Parsed into minutes, including overnight spans; missing hours set to the median and flagged	weekly_hours, opens_early, open_late, has_hours
Website, menu, ordering links	Own sites mixed with Instagram or Facebook pages	Domain check; presence of menu and ordering-partner links	own_website, social_page_only, has_menu_link, online_ordering
Owner and phone	Personal data	Removed; only presence kept	is_claimed, has_phone
Coordinates	Raw latitude and longitude	Haversine distance to Town Hall; count of other venues within 1 km	dist_centre_km, log_density
Review count	Highly skewed (10 to 36,297)	$\log(1 + x)$	log_reviews

Leakage control. The 1- to 5-star counts, review text and Google *Highlights* (such as “Great coffee”, which are derived from reviews) were excluded because they are built from the rating itself. *Popular for*, *Crowd* and *Atmosphere* were excluded because they describe customers rather than decisions of the owner. The largest variance inflation factor is 3.75, so multicollinearity is not a concern.

3.2 Exploratory data analysis

Restaurant type and price matter. Cafés and beverage outlets reach the benchmark in 78.3% of cases, followed by fast food (68.3%) and bakeries (66.3%), while traditional South Indian restaurants (32.2%), biryani and non-vegetarian venues (41.8%) and vegetarian restaurants (43.2%) have the lowest rates ($\chi^2 = 241.7$, $p < 0.001$; Figure 4a). These value-driven formats serve large volumes and are judged against strong home-cooking expectations. Price follows a non-linear pattern ($\chi^2 = 40.5$, $p < 0.001$; Figure 4b): premium venues perform best (71.4%) and the mid-price band of ₹200–400 performs worst (48.1%).

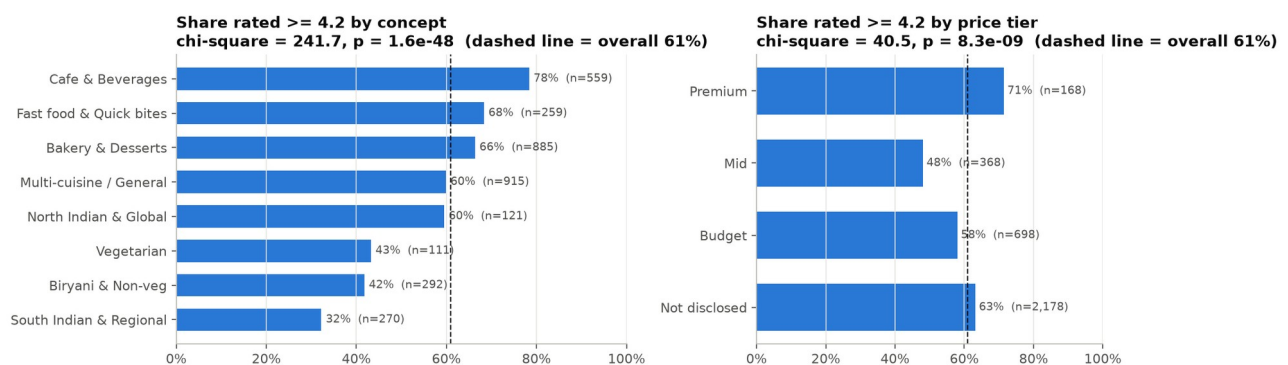


Figure 4: Share of venues rated 4.2 or higher (a) by restaurant type and (b) by price tier; n per group in brackets.

Profile completeness separates venues. Figure 5 compares the success rate with and without each yes/no attribute. The largest gap for a common attribute is the **claimed Google profile**: 67.3% of claimed venues reach 4.2 against 42.9% of unclaimed ones, and 26% of venues have not claimed their profile. A website, menu link and online-ordering link show similar positive gaps. Wi-Fi shows the largest gap (85.2% against 60.1%) but is offered by only 3.4% of venues. Table service, card payments and restrooms show negative gaps; these attributes mark formal, full-service restaurants and are examined further in the multivariable model.

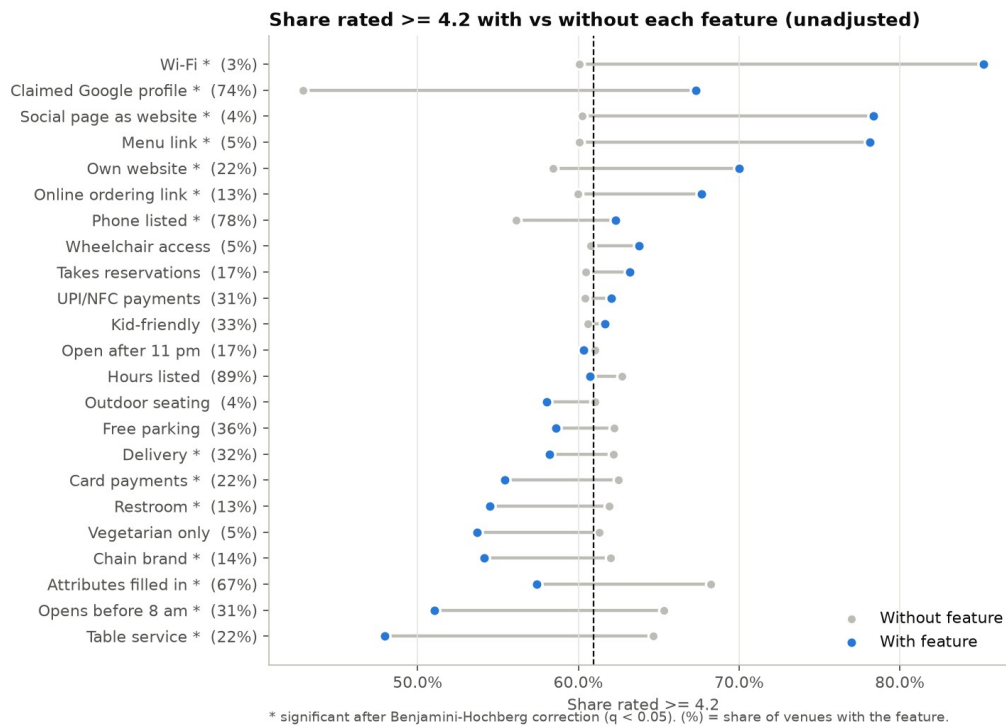


Figure 5: Share of venues rated 4.2 or higher with and without each attribute (unadjusted); * significant after Benjamini-Hochberg correction; percentages show how common each attribute is.

Long operating days go with lower ratings. Venues open less than 60 hours a week reach the benchmark in 84.5% of cases, against 44.2% for those open 120 hours or more, mostly tea shops, messes and 24-hour outlets (Figure 6a). Opening before 8 am looks negative on its own (51.0% against 65.3%; Figure 6b), but the model in Section 6 shows that, for the same total hours, early and late opening are both positive.

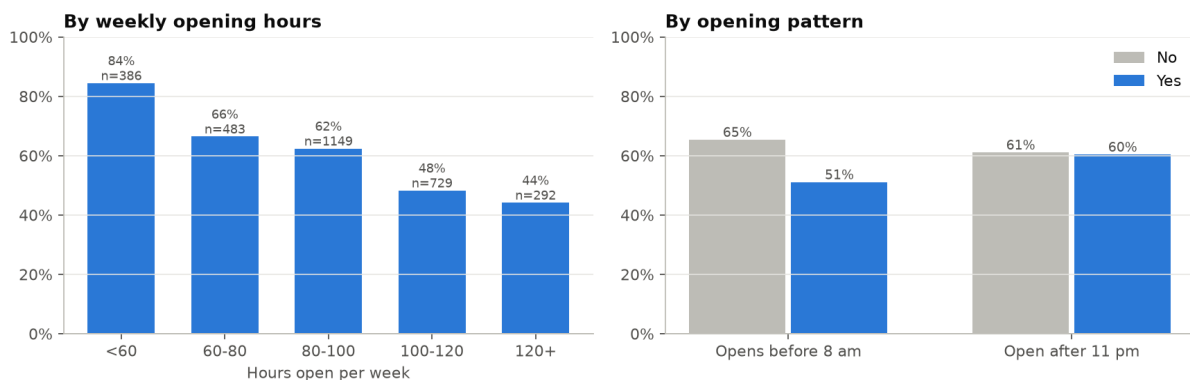


Figure 6: Share of venues rated 4.2 or higher (a) by weekly opening hours and (b) by early and late opening.

Location has a smaller effect. The success rate peaks 2–4 km from the city centre (66.0%) and falls to 53.1% beyond 20 km (Figure 7). No single feature is strongly correlated with the target; the largest correlations are the claimed profile ($r = 0.22$), weekly hours ($r = -0.21$) and review volume ($r = -0.17$), so success depends on many small factors acting together, which supports a multivariable model.

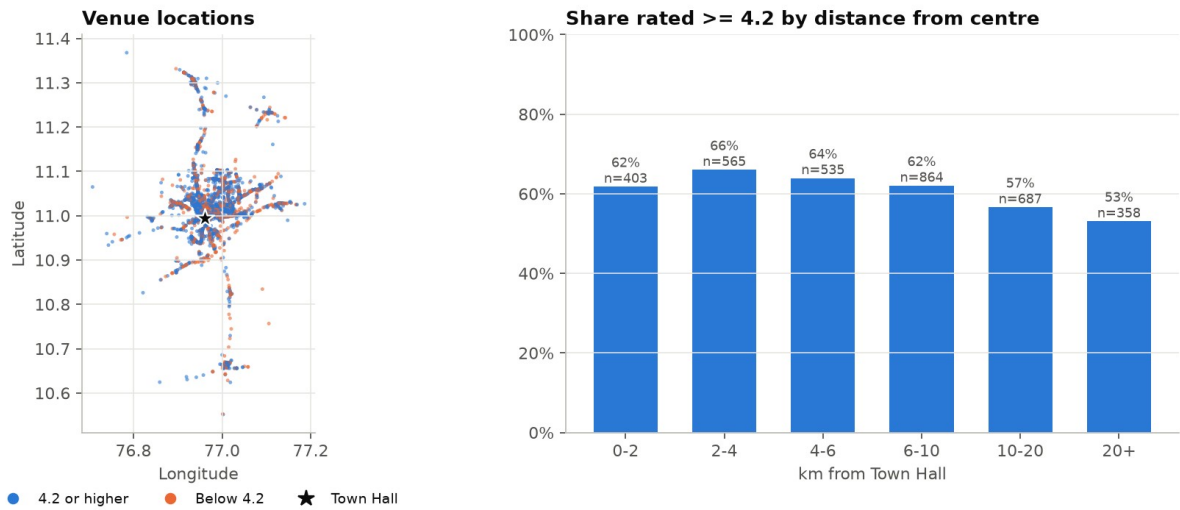


Figure 7: (a) Venue locations coloured by rating class; (b) share of venues rated 4.2 or higher by distance from Town Hall.

4. Analytics Method and Implementation

The task is **binary supervised classification** (4.2 or higher = 1, below 4.2 = 0). Three standard classification techniques [6] were compared: **Logistic Regression (LR)**, which models the log-odds of reaching the benchmark as a linear function of the features, $P(y = 1 \mid \mathbf{x}) = 1 / (1 + e^{-(\beta_0 + \sum \beta_i x_i)})$, and gives odds ratios that can be read directly as business effects; the **Decision Tree (DT)**, which produces if-then rules from Gini-based splits; and **K-Nearest Neighbours (KNN)**, which classifies a venue from the most similar venues and assumes no functional form. A **Random Forest (RF)** of 400 trees was added as a benchmark to test whether a more flexible ensemble improves prediction. Interpretability was a requirement, since a recommendation is only useful if an owner can see why it is made.

Implementation (pandas [7], scikit-learn [8] Pipelines and statsmodels [9]; scaling is fitted on training data only):

- **Features:** 37 model columns, with *Multi-cuisine / General* and *Not disclosed* as reference levels. Identifiers, the rating and the star counts were excluded, and a check in the code stops the analysis if any rating-derived column enters the feature matrix.
- **Split and scaling:** stratified 80/20 split (2,729 training and 683 test venues); continuous features standardised for LR and KNN; `class_weight = "balanced"` for LR, DT and RF to handle the 61/39 class split.
- **Tuning and selection:** 5-fold stratified grid search on the training set, scored by ROC-AUC – LR: $C \in \{0.01, 0.1, 1, 10\} \rightarrow 0.1$; DT: depth $\{3, 4, 5, 6, 8\} \times$ minimum leaf $\{10, 25, 50\} \rightarrow$ depth 8, leaf 50; KNN: $k \in \{5, 11, 21, 31, 51\} \times$ uniform/distance weights $\rightarrow k = 51$, distance; RF: depth $\{6, 10, \text{None}\} \times$ leaf $\{5, 10, 20\} \times$ features $\{\sqrt{p}, 0.3\} \rightarrow$ unlimited depth, leaf 5, $\sqrt{p}$. The test set was used once.
- **Evaluation and interpretation:** accuracy, precision, recall, F1 and ROC-AUC, with ROC-AUC as the main metric because it does not depend on a single threshold; 5-fold cross-validation on the full sample for stability; a statsmodels logistic regression for odds ratios with 95% confidence intervals and average marginal effects (change in probability, in percentage points); and RF permutation importance as a non-linear check.

5. Comparison with State-of-the-Art Methods

Table 4: Comparison of this case study with recent published studies.

Published study / year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
Zhang and Luo [10], Management Science 2023	Yelp: 17,719 U.S. restaurants, 755,758 photos, 1.12 M reviews (2004–2015)	Deep-learning photo features + XGBoost; causal forests for interpretation	Out-of-sample prediction of restaurant survival	Photos predict survival up to three years ahead, reviews about one year; share of food photos is the strongest signal	Same idea of predicting outcomes from platform data, but with photos and a survival target; this study uses structured listing attributes and a rating benchmark.
Li, Yu, Li and Gao [11], Tourism Management 2023	Yelp reviews of restaurants in two major U.S. tourist destinations	Aspect-based sentiment (location, taste, price, service, atmosphere) + conditional survival forest	Survival prediction vs an overall-sentiment baseline	Aspect-level sentiment predicts survival better than overall sentiment	Shows the value of review content; review text is excluded here to avoid leaking the rating, so the levers apply before reviews exist.
Li, Zhou, Zheng, Xu and Zhang [12], Tourism Management 2025	Yelp: 2,838 Boston restaurants, 528,592 reviews (2004–2021)	Bayes, SVM, MLP and XGBoost with review-variance features	One-year survival prediction performance	Variance of review features predicts better than the average rating; chain affiliation, price level and age are key determinants	Agrees that chain status and price matter; this study adds profile, service and opening-hours attributes in an Indian market.
Starakiewicz and Wójcik [13], Knowledge-Based Systems 2025	Nationwide Google Maps listings, Poland (collected 2022)	Label-free semantic review features, clustering-based feature selection, ML classifiers	ROC-AUC on unseen data	Baseline model ROC-AUC above 0.7 for predicting restaurant exit	Closest data source (Google Maps, binary target, ROC-AUC); the 0.77–0.78 here is in a similar range, but targets and markets differ.

The published studies use larger datasets from Yelp or a nationwide Google Maps scrape, and they predict restaurant survival rather than rating level. Because datasets, targets and markets differ, **the reported scores are not directly comparable and no method is claimed superior on the basis of its score**. At the level of findings, all studies agree that information visible on review platforms carries real signal about a restaurant's prospects, and [12] also identifies chain affiliation and price level as important, consistent with the chain and mid-price results here. The distinct contributions of this study are its focus on attributes an owner can change, the exclusion of review content to prevent target leakage, and the local Indian setting; its main limitations are the single snapshot in time and the absence of review text and photographs.

6. Results, Business Insights and Recommendations

Table 5: Five-fold cross-validation and hold-out test performance (threshold 0.5).

Model	CV ROC-AUC (mean $\pm$ SD)	CV F1	Test accuracy	Precision	Recall	F1-score	Test ROC-AUC
Logistic Regression	0.768 $\pm$ 0.030	0.745	0.688	0.787	0.668	0.723	0.759
Decision Tree	0.718 $\pm$ 0.024	0.709	0.666	0.755	0.668	0.709	0.733
K-Nearest Neighbours	0.760 $\pm$ 0.028	0.787	0.737	0.737	0.882	0.803	0.763
Random Forest (benchmark)	0.783 $\pm$ 0.028	0.761	0.726	0.794	0.743	0.768	0.784
Majority-class baseline	0.500	–	0.609	0.609	1.000	0.757	0.500

All four models clearly outperform the majority-class baseline. Among the three main models, **Logistic Regression ranks venues best** (CV ROC-AUC 0.768 $\pm$ 0.030) and has the highest precision. KNN records the highest F1 (0.787), but mainly because it predicts the benchmark for almost every venue (recall 0.863); with 61% positives this inflates F1, which is why ROC-AUC is used for selection. The Decision Tree is the weakest (0.718). The Random Forest benchmark reaches the best CV ROC-AUC (0.783), 0.015 above LR and within the

fold-to-fold variation. Test scores match the cross-validation results, so the models do not overfit. Random Forest is therefore preferred for prediction, while Logistic Regression is used to explain the factors. A ROC-AUC near 0.78 is reasonable for listing data, which contains no direct measure of food quality or service.

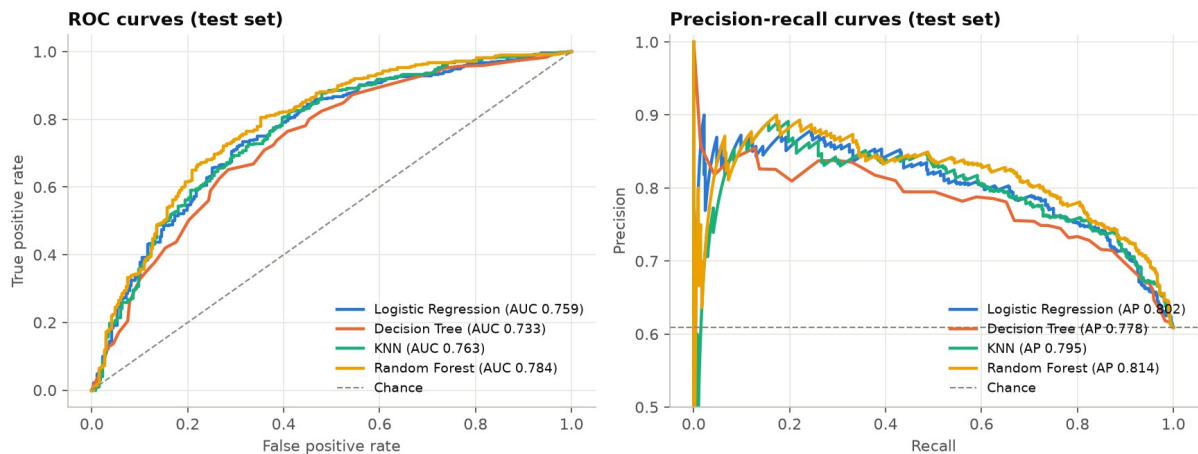


Figure 8: (a) ROC curves and (b) precision-recall curves of the four models on the test set (n = 683).

Factors associated with the 4.2 benchmark

The logistic regression is highly significant (McFadden pseudo- $R^2 = 0.184$; likelihood-ratio test $p < 0.001$). Holding the other features fixed (Figure 9a):

- a **claimed Google profile** multiplies the odds of reaching 4.2 by **2.22** (95% CI 1.83–2.69), equal to **+15.5 percentage points**;
- an **own website** (OR 1.65), **menu link** (1.86), **online-ordering link** (1.62) and **UPI/NFC payments** (1.53) are also positive; Wi-Fi has the largest odds ratio (3.61) but a wide interval, as only 3.4% of venues offer it;
- cafés (2.50) and bakeries (1.81) have higher odds than multi-cuisine restaurants, South Indian (0.38) and biryani/non-vegetarian venues (0.51) lower odds, and chain outlets lower odds than independents (0.61);
- each additional standard deviation of weekly hours (about 27 hours) halves the odds (0.50), while for the same total hours opening before 8 am (1.45) and after 11 pm (1.56) are positive;
- the mid-price band is the only significant price tier (0.57); table service (0.53) and card payments (0.59) are negative markers of formal restaurants rather than practices to remove.

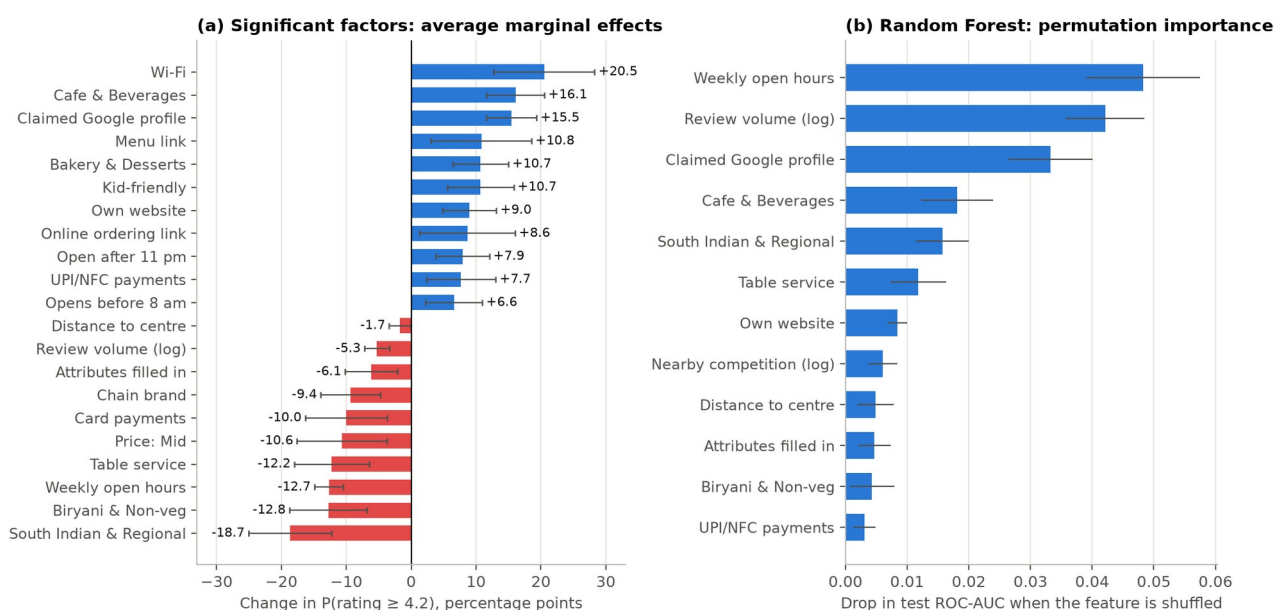


Figure 9: (a) Average marginal effects of the significant features with 95% CIs, in percentage points; (b) Random Forest permutation importance on the test set.

The Random Forest ranks the same drivers highest – weekly hours, review volume, the claimed profile and the café and South Indian types (Figure 9b) – so the result does not depend on the linear form of the model. A depth-3 decision tree turns the findings into simple rules (Table 6): claimed venues open no more than about 105 hours a week that are not biryani outlets, half of the sample, reach 4.2 in 75% of cases, whereas unclaimed venues with more than 37 reviews located over 4.5 km from the centre reach it in only 25%.

Table 6: Rules from a depth-3 decision tree fitted on the full modelling sample.

Rule	Share of venues	Reach 4.2+
Not claimed; up to 37 reviews; open up to 78 h per week	2.5%	78.6%
Claimed; open up to 105 h per week; not biryani/non-veg	50.6%	75.0%
Claimed; open more than 105 h per week; up to 248 reviews	11.0%	61.3%
Not claimed; up to 37 reviews; open more than 78 h per week	8.0%	54.4%
Claimed; open up to 105 h per week; biryani/non-veg	5.2%	48.0%
Not claimed; more than 37 reviews; within 4.5 km of the centre	5.3%	44.0%
Claimed; open more than 105 h per week; more than 248 reviews	7.0%	35.8%
Not claimed; more than 37 reviews; beyond 4.5 km of the centre	10.3%	24.8%

Robustness. The conclusions hold under several checks. Re-estimating with minimum review counts of 5, 25 and 50 keeps every significant odds ratio on the same side of 1 (the claimed-profile OR stays between 2.18 and 2.39); removing the review-count control does not change the direction of any key factor; and an OLS regression on the continuous rating ($R^2 = 0.25$) agrees in direction on all 21 significant factors, 20 of which are also significant. A further test with five features built from Google's popular-times data (available for 35% of venues) lowered cross-validated ROC-AUC for every model (by 0.001 to 0.014), so these features were not used.

Recommended actions

The findings are translated into eight actions, ordered by strength of evidence and cost (Table 7). The expected effects are model-based estimates for venues that do not yet have the attribute; because the data are observational, they indicate direction and rough size rather than guaranteed outcomes.

Table 7: Prioritised actions for restaurant owners.

#	Action	Supporting evidence	Effort and cost
1	Claim the Google Business Profile, keep details current and reply to reviews	67% of claimed venues reach 4.2 against 43%; OR 2.22; for the 891 unclaimed venues the predicted chance rises from 43% to 59%	Low; free
2	Accept UPI/NFC payments and show them on the profile	OR 1.53; about +8 percentage points	Low; almost no cost
3	Complete the online storefront: own website or menu link, and an aggregator ordering link	OR 1.62–1.86; about +9 to +11 percentage points each	Low to medium
4	Avoid very long operating days; add a focused breakfast or late-night window instead	Odds halve per 27 extra weekly hours; early and late slots are positive at the same total hours	Medium; staffing
5	Position the price clearly: compete on value or move to a distinct premium offer	Mid-price venues reach 4.2 in 48% of cases (OR 0.57)	Medium
6	In South Indian and biryani venues, prioritise consistency of taste, hygiene and speed	Success rates of 32% and 42%, the lowest of all restaurant types	Medium
7	For chains, audit outlet-level consistency against the brand standard	Chain outlets have lower odds than independents (OR 0.61)	Medium
8	Offer Wi-Fi where the format suits longer stays, such as cafés	OR 3.61, but only 3.4% of venues offer it, so the evidence is limited	Low

7. Conclusion

The study applied the full business-analytics workflow to the question of what separates Coimbatore food venues that reach the 4.2-star premium benchmark. Through 1,200 Google Maps searches, 4,917 listings were web-scraped, cleaned into a dataset of 4,667 food venues and analysed on 3,412 reliably rated venues. The first objective was met with an original, anonymised dataset built entirely by web scraping. For the second, Logistic Regression predicted the benchmark from listing data alone with a cross-validated ROC-AUC of 0.768, and a Random Forest benchmark reached 0.783. For the third, a claimed and complete Google profile, digital payments and focused operating hours emerged as the strongest attributes under an owner's control, and were turned into eight prioritised actions.

Limitations and future work

- The data are observational and collected at one point in time, so the effects are associations rather than proof of cause; repeating the scrape over time would show whether ratings change after a venue adopts an action.
- Some attributes are contributed by users and may be incomplete, and food quality and service, the main drivers of ratings, are not observed; review-text sentiment and venue age are the most useful additions.
- The study covers one district; extending it to other cities in Tamil Nadu would test whether the findings generalise.

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